Operating Systems Examination Paper

2024 Examination

Unit I - Processes

Q1. Explain process states and the process control block (PCB).

Unit II - Memory Management

Q2. Explain paging, pages, and frames with architecture diagrams.

Q3. Explain LRU page replacement algorithm with an example page reference string.

Q4. Discuss file allocation methods including contiguous, linked, and indexed.

2023 Examination

Unit I - Processes

Q1. Describe process states and the process control block (PCB) structures.

Q2. Explain paging / pages / frames memory translation mechanism.

Unit II - Storage & Concurrency

Q3. Explain virtual memory concepts and demand paging techniques.

Q4. Describe deadlock conditions and the four necessary Coffman conditions.

2022 Examination

Unit I - Memory

Q1. Explain LRU page replacement and evaluate its fault rate.

Q2. Discuss virtual memory management and page fault handling.

Unit II - File Systems & Deadlocks

Q3. Explain file allocation schemes and compare contiguous vs indexed.

Q4. Discuss deadlock conditions and avoidance using Banker’s algorithm.